Range-wide population genetics reveals how disjunctions, admixture, and translocations influence conservation of *Physaria globosa* (Brassicaceae)

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Abstract

Population genetic analyses of endangered species can directly inform management strategies to conserve genetic diversity and increase long-term survival. *Physaria globosa*, an endangered mustard, has a geographically restricted and disjunct distribution across river systems in Indiana, Kentucky, and Tennessee, USA. In this study, we assessed whether genetic structure was shaped by the species' disjunct distribution, whether genetic diversity has been affected by bottlenecks, drift, or inbreeding, and whether translocations based on mixed sources can retain genetic variation. We genotyped 332 individuals at 20 microsatellite loci from 14 natural and 2 translocated populations from across the species' range. The genetic structure of natural populations showed two major geographic groups with little to no connectivity between them: a northern group in Indiana and Kentucky (IN-KY) and a southern group in Tennessee (TNE-TNW). Genetic diversity was highly variable among and within geographic regions; however, several local populations showed reduced genetic diversity, inbreeding, or recent bottlenecks, consistent with demographic decline and ongoing pressures from habitat degradation and competition from invasive plants. The two translocated populations showed clear genetic mixing of source populations but showed depletion of private alleles. Our results suggest that conservation actions for *P. globosa* should prioritize the protection of multiple populations within each major geographic region. Combining material from mixed sources appears to be a viable approach to increase genetic diversity and restore gene flow, but we recommend that translocations combine material from within the same geographic regions to preserve the distinct evolutionary histories of the main geographic groups.

Keywords: Genetic structure; genetic mixing; conservation management; microsatellites; translocation

Introduction

Genetic diversity is widely recognized as the raw material underlying phenotypic and functional variation within species, and its maintenance is a central goal of biodiversity conservation (Shaw et al. 2025). Greater genetic diversity is often associated with greater resilience to environmental change, and increased evolutionary potential and individual fitness. Genetic diversity can also improve the establishment and long-term persistence of reintroduced populations (Hoelzel, Bruford, & Fleischer 2019; Van Rossum & Le Pajolec 2021). For endangered species with narrow distributions and ongoing habitat loss and degradation, conserving genetic diversity requires consideration of natural populations, *ex situ* collections, and conservation translocations derived from them (Aguilar-Gómez et al. 2025; Edwards et al. 2025; Linan et al. 2024; Mashburn et al. 2023; Van Rossum et al. 2025).

Conservation strategies for rare plant species generally span a continuum from protecting extant natural populations to more interventionist approaches involving the establishment of *ex situ* populations and translocations into suitable habitats (Bucharova 2017; Gaitán-Espitia & Hobday 2021). In plants, these interventions can include controlled crosses among natural populations and translocation of individuals among populations to promote interbreeding and restore genetic variation (Edwards et al. 2025). Such actions can be valuable when fragmentation, demographic decline, or habitat degradation have reduced effective population sizes and limited natural gene flow. In these circumstances, facilitated gene flow can

increase genetic diversity, reduce inbreeding, and potentially enhance fitness and longer-term adaptive potential. However, outcomes are often context-dependent and not always predictable; thus, their application remains debated (MacDonald et al. 2025).

Using mixed-source materials for controlled crosses, population augmentations, translocations, and other forms of assisted migration raise several important questions in conservation genetics. Do disjunct populations represent distinct historical or adaptive components that should be protected separately (Fraser & Bernatchez 2001)? Under what conditions does the risk of loss of genetic diversity due to inbreeding or genetic drift in an isolated population exceed the risk associated with admixture? Can mixed populations capture sufficient genetic diversity to maintain their persistence in local or changing environments? The major concerns are well known: introduced gene flow may cause outbreeding depression, disrupt local adaptation, generate maladapted genotypes, or generate ecological mismatches in recipient environments (Bucharova 2017; Hufford & Mazer 2003; MacDonald et al. 2025). Hybridization can also lead to genetic or demographic swamping if rare lineages are overwhelmed by introgression from more widespread relatives (Aguilar-Gómez et al. 2025). In addition, small, inbred but viable populations may have already purged some deleterious recessive alleles, such that genetic rescue could introduce new genetic load and reduce fitness (Dussex, Morales, Grossen, Dalen, & van Oosterhout 2023). These risks do not preclude intervention but underscore the need to identify a clear biological and demographic rationale, carefully evaluate risks of both source and recipient populations, and conduct post-release genetic monitoring to assess outcomes (Aguilar-Gómez et al. 2025; Linan et al. 2024).

Globe bladderpod, *Physaria globosa* (Desv.) O'Kane & Al-Shehbaz, is a federally endangered annual to biennial herbaceous mustard with a geographically restricted and discontinuous distribution in the southeastern United States (Fuentes-Soriano & Kellogg 2021). It occurs on steep, rocky bluffs and talus slopes, usually on calcareous substrates with south- to west-facing slopes near rivers or streams (Fig. 1A). The species is distributed across four disjunct regions: the confluence of the Wabash and Ohio Rivers in Indiana (IN), the Kentucky River watershed in Kentucky (KY), and the Cumberland River watershed in Tennessee (TN), which is separated into eastern (TNE) and western (TNW) groups, divided by the Nashville metropolitan area (Fig. 2). *Physaria globosa* is threatened by habitat loss and degradation, soil erosion, and woody encroachment, particularly from invasive nonnative shrubs (Kallison et al. 2026; Long, Albrecht, Call, & Region 2021). Despite its conservation status, it remains unclear how genetic variation is distributed across these disjunct regions and whether individual populations are experiencing genetic risks such as bottlenecks, inbreeding, or genetic erosion.

Climate change may further intensify these threats, as extended summer droughts and extreme precipitation events can increase both physiological stress and flood- or erosion-related habitat disturbance (Donat, Lowry, Alexander, O'Gorman, & Maher 2016; Xu et al. 2019). Together, these ecological and genetic vulnerabilities make *P. globosa* a useful system for evaluating whether conservation efforts should prioritize conserving existing population structure in natural populations and *ex situ* conservation collections or restoring genetic diversity through translocation using mixed-source material to facilitate admixture. Although such interventions may help maintain genetic diversity and promote long-term persistence, they also carry risks, including outbreeding depression, disruption of adaptive differentiation,

and incomplete representation of natural genetic diversity in *ex situ* collections. Management decisions should therefore be guided by range-wide data on genetic diversity and population structure, while also evaluating the consequences of using multiple sources for conservation translocations.

This study addresses these questions using a comprehensive range-wide analysis of genetic diversity and structure in *P. globosa*. We sampled 14 natural populations spanning the species' distribution and two mixed-source translocated populations. We genotyped 332 individuals at 20 microsatellite loci. Our goals were to understand: 1) whether genetic structure was shaped by the species' disjunct distribution, 2) whether genetic diversity in populations has been affected by bottlenecks, drift, or inbreeding; and 3) whether existing mixed-source translocations retain genetic variation while remaining consistent with the broader recovery goal of promoting representation, resiliency, and redundancy across the range of this endangered species. Finally, we evaluated the conservation implications of these patterns for both *in situ* protection and *ex situ* management.

Materials and Methods

Study species

Physaria globosa is a member of the mustard family (Brassicaceae). It is morphologically distinct from other eastern and south-central U.S. congeners, including *P. ouachitensis* and *P. filiformis*, which are annual plants with fibrous or slender root systems, whereas *P. globosa* is most frequently biennial or sometimes annual, and often develops a woody caudex (Flora of North America, *Flora of North America* 2010; Fuentes-Soriano & Al-Shehbaz 2013; Fuentes-Soriano & Kellogg 2021; Kallison et al. 2026). Plants may produce multiple stems and grow up

to 50 cm tall. Small yellow flowers are produced in April and May and are pollinated primarily by small native bees and flies (Krosnick et al. 2022). In June and July, the species produces distinctive globular fruits that give bladderpods their common name. Seeds are thought to be dispersed by water and gravity along the rivers (Long et al. 2021), or transported by small mammals (Tara Littlefield, pers. observation). We have observed that *Physaria globosa* seeds can persist in soil for at least five years (unpublished data), which is common in the genus *Physaria* (Elberling 2000).

Sample collection and DNA extraction

We sampled *Physaria globosa* across its known geographic range in May 2021, with sampling effort in each region proportional to the number of known occurrences in each of the four main geographic regions (**Table 1**). In total, we collected 332 individuals from 16 populations. These included the only known natural population in Indiana, USA (IN_01); seven natural populations in Tennessee, USA, including two eastern Tennessee populations (TNE_01, TNE_02) and five western Tennessee populations (TNW_01, TNW_02, TNW_03, TNW_04, TNW_05); six natural populations in Kentucky, USA (KY_01, KY_02, KY_03, KY_04, KY_05, KY_06); and two translocated populations in Kentucky, USA (TR_01, TR_02).

The two translocated populations had distinct establishment histories. TR_01 was established by Tara Littlefield using two sources of introduced material. The first source was from a road-cut rescue at Cove Spring, Kentucky, USA (KY_02) in 2014. Approximately 100 seeds from 30-40 maternal plants were collected and moved to a suitable habitat on the east end of Cove Spring Park, approximately one mile away from the original site. The introduction site had recently been cleared of honeysuckle prior to seed addition, within an approximately 5

× 5 m managed area. The habitat included limestone rock outcrops and associated woody species such as *Juniperus virginiana*, *Quercus muehlenbergii*, *Fraxinus quadrangulata*, *Cercis canadensis*, and *Ostrya virginiana*. The second source consisted of seeds collected from Woodford County, Kentucky, USA (KY_05) in 2015. Approximately 200 additional seeds from 5-6 maternal plants were introduced to the site. Honeysuckle removal continued in winter 2015, with management focused on expanding the original cleared area where *Physaria globosa* was introduced. Therefore, the TR_01 population could be described as a field-established introduction with mixed seed-source origins. TR_01 has been monitored annually by Tara Littlefield, local volunteers, and staff from the Office of Kentucky Nature Preserves. No plants were observed inside the original 5 × 5 m managed area in 2014, followed by two vegetative plants in 2015, six vegetative and one flowering plant in 2016, with continued annual increases in flowering plants through 2025. Later increases appear to coincide with continued habitat management and honeysuckle removal at the site. The TR_01 population reached 238 vegetative and 29 flowering plants in 2025.

The TR_02 population at Jessamine Creek Gorge Preserve was established by a local conservation enthusiast. To the best of our knowledge, plants were originally sampled from populations in both Kentucky and Tennessee in the early 1990s and established on private property in a mixed-source population. These plants were allowed to naturally cross-pollinate. Seeds resulting from the mixed-source population were harvested and were broadcast-sown approximately three times at Jessamine Creek Gorge Preserve from 2010 to 2020. The specific source locations for the original population and the number of seeds used in the translocation were not recorded.

For both wild and translocated populations, we sampled leaf tissue from 10 -24 individuals per population, depending on population size and site accessibility. We also collected one herbarium voucher specimen per population, which was deposited in the Missouri Botanical Garden Herbarium (MO). The sampling locations, adjacent rivers, main cities, and human footprint (Gassert et al. 2023) were plotted in **Fig. 1**. To provide genus-level context for genetic diversity estimates, we also included 17 individuals from three populations of *P. filiformis* (Rollins) O'kane and Al-Shehbaz in Missouri, USA, 19 individuals from three populations of *P. ouachitensis* Kallison and C.E. Edwards in the Ouachita Mountains of Arkansas, USA, and 12 individuals from one population of *P. gracilis* (Hook.) O'kane and Al-Shehbaz in Texas, USA (**Table 2**), all previously included in C. E. Edwards et al. (2021). The final dataset comprised 406 individuals. Leaf tissue was dried in silica gel in the field, and DNA was extracted from silica-dried tissue using a modified CTAB protocol (Allen, Flores-Vergara, Krasynanski, Kumar, & Thompson 2006).

Microsatellite markers and genotyping

We developed and employed 20 microsatellite markers to genotype *P. globosa*. We first tested 16 microsatellite markers previously developed for *Physaria filiformis* (C. E. Edwards et al. 2021), of which 10 produced polymorphic PCR bands of the expected size in *P. globosa*. We then developed 10 additional microsatellite markers for *P. globosa* using the protocol described in Swift, Smith, Menges, Bassuner, and Edwards (2016). Briefly, whole-genome libraries were prepared for two *P. globosa* individuals, one from Tennessee and one from Kentucky, using Nextera DNA Sample Prep Kits. Libraries were sequenced on an Illumina MiSeq platform with 2 × 150 bp reads. Sequencing yielded 1,049,048 and 1,163,933 paired-end reads, respectively,

199 which were assembled de novo into 585,371 contigs using the medium sensitivity/fast setting in
200 Geneious v.7.1.9 (Biomatters, Auckland, New Zealand). MSATCOMMANDER (Faircloth 2008)
201 identified 2,054 unique microsatellites from the contig consensus sequences, and PRIMER3
202 (Rozen & Skaletsky 1999) identified 324 unique primer pairs. An M13 tag (5'-
203 CACGACGTTGTAAAACGAC-3') was added to the 5' end of each forward primer to enable
204 universal dye labeling (Boutin-Ganache, Raposo, Raymond, & Deschepper 2001). Of 48 primer
205 pairs tested, 10 additional polymorphic loci showed reliable amplification in *P. globosa*. These
206 20 loci (**Appendix 1, Table 1**) were used to genotype the 406 samples.
207 PCR amplifications were performed in 10 µL reactions containing 0.5 U GoTaq Flexi DNA
208 polymerase (Promega, Madison, WI, USA), 1× Colorless GoTaq Flexi Buffer (Promega, Madison,
209 WI, USA), 1.5 mM MgCl2, 4.5 pmol each of the reverse primer and one of four fluorescently
210 labeled M13 primers (6-FAM, VIC, NED, or PET; Applied Biosystems, Foster City, CA, USA), 0.18
211 pmol of the M13-tagged forward primer, and 0.5 mM of each dNTP. Thermal cycling consisted
212 of an initial denaturation at 94 °C for 3 min; 35 cycles of 94 °C for 30 s, 52 °C for 30 s, and 72 °C
213 for 45 s; and a final extension at 72 °C for 20 min. Genotyping was conducted by capillary
214 electrophoresis on an Applied Biosystems 3730xl (Applied Biosystems, Foster City, CA, USA) at
215 the Cornell Biotechnology Resource Center. Raw trace files were scored in GeneMarker v.2.6.2
216 (SoftGenetics, State College, PA, USA).

217 Analysis of population structure

218 Population genetic structure was characterized by Nei's standard genetic distance
219 among populations, pairwise F_{ST}, and pairwise G_{ST}, calculated in Genodive (Meirmans & Van
220 Tienderen 2004). The Nei's *D* matrix was used to construct a neighbor-joining tree in R using the

ape package (Paradis, Claude, & Strimmer 2004), rooted with *P. gracilis* as the outgroup. Genetic structure within *P. globosa* was also examined by PCA in the R package `adegenet` (Jombart 2008) and by Bayesian clustering in STRUCTURE v.2.3.4 (Pritchard, Stephens, & Donnelly 2000). We ran five replicate analyses for each *K* from 1 to 13, using a burn-in of 600,000 generations and a run length of 1,000,000 generations, and evaluated the most likely number of clusters in STRUCTURE HARVESTER (Earl & vonHoldt 2012). The result was visualized on a map using the R package scatterpie (Yu 2025). Hierarchical genetic structure was further quantified by AMOVA in GenAlEx (Peakall & Smouse 2012). We explored three population grouping strategies in AMOVA to determine which best explained patterns of among-population genetic structure, including two groups (IN+KY and TNE and TNW), three groups (IN+KY, TNE, and TNW), and four groups, with each ecoregion in a separate group.

Analysis of genetic diversity

Samples with >50% missing data were excluded, removing nine *P. globosa* individuals and leaving a final dataset of 397 individuals. The mean number of alleles per population (A), observed and expected heterozygosity (H_O, H_E), and the inbreeding coefficient (F_{IS}) were calculated by Genodive (Meirmans & Van Tienderen 2004). Private alleles (A_P) were identified by GenAlEx (Peakall & Smouse 2012), and allelic richness (A_R) was estimated by HP-Rare (Kalinowski 2005). Null allele frequency (N), genotyping error (b), and the inbreeding coefficient corrected for null alleles (F_B) were estimated by INEST v.2.2 (Chybicki & Burczyk 2009) under a Bayesian MCMC framework with 5,000,000 cycles, sampling every 250 cycles after a burn-in of 500,000. Recent bottlenecks were assessed by the bottleneck function in INEST v.2.2 based on heterozygosity excess and M-ratio deficiency, each evaluated with a Z-test and a Wilcoxon

signed-rank test. Populations were considered to show evidence of a recent bottleneck when at least three of the four tests were significant ($p < 0.05$).

Results

Interspecific and intraspecific differentiations

We sampled *P. globosa* from one natural population in the Indiana region (IN), six natural populations in the Kentucky region (KY), five natural populations in western Tennessee (TNW), two natural populations in eastern Tennessee (TNE), and two translocated populations (**Fig. 1A, B; Fig. 2**). Broad-scale population structure was consistent across the neighbor-joining tree, STRUCTURE, PCA, and pairwise differentiation analyses. The neighbor-joining tree (**Fig. 1C**), rooted with *P. gracilis*, placed the three *P. filiformis* populations in a group with *P. gracilis*, while the three *P. ouachitensis* populations formed a sister group to *P. globosa*. Within *P. globosa*, populations were divided into 2 major groups: an IN–KY group and a TNE–TNW group. Pairwise F_{ST} and G_{ST} showed a similar pattern (**Fig. 3C**), further supporting this major split among natural populations. Consistent with this result, PCA separated IN–KY populations from TNE–TNW populations along PC1, which explained 16.7% of the variance (**Fig. 3B**).

STRUCTURE analyses provided additional resolution of this pattern. Across analyses of K = 1–13, the Evanno method identified K = 2 as the best-supported number of clusters. At K = 2, the IN and KY populations formed a genetic cluster and were separated from the 2nd genetic cluster composed of TNE and TNW populations (**Fig. 3A**). Exploration of higher K values revealed finer-scale structure. At K = 3, a gradient-like pattern was evident among the TNE and TNW populations, except for the TNW_04 and TNW_05, which remained genetically distinct

and largely unadmixed through the highest *K* examined. The Nashville (TN) cosmopolitan area did not block the natural gene flow between the TNE and TNW populations (**Fig. 2**). Instead, the genetic distinctiveness of the TNW_04 and TNW_05 populations could be a result of elevated inbreeding (**Fig. 4D**). At *K* = 5, the IN population formed its own genetic cluster, although it remained more similar to the KY than to the TNE and TNW populations, and a gradient-like pattern was also visible among the KY populations (**Fig. 3A**). PCA supported these finer-scale results: PC2, which explained 6.7% of the variance, further separated TNE and TNW populations, except for TNW_03, which clustered with the TNE populations (**Fig. 3B**).

The translocated populations also showed genetic patterns consistent with their documented origins. The TR_01 clustered with the KY populations in both STRUCTURE and PCA, consistent with its origin from the KY_02 and KY_05 populations (**Fig. 3A, B**). At K = 4-5, four individuals (16.7%) of the TR_01 population showed potential admixture between the two source populations, 16 individuals (66.7%) grouped with KY_02 and four individuals (16.7%) grouped with KY_05 (**Fig. 3A**). The KY_02 population had more genetic contributions to the TR_01 population than the KY_05 population (**Fig. 3C**). The TR_02 population was intermediate between the IN-KY and TNE-TNW populations in PCA and showed admixed ancestry in STRUCTURE (**Fig. 3A, B**). At *K* = 5, STRUCTURE indicated that the TR_02 population included individuals with varying degrees of admixture, including putative F_1 hybrids and backcrosses (**Fig. 3A**). In addition, G_{ST} indicated that the TR_02 population received stronger genetic contributions from the TNW_04 and TNW_05 populations than from other Tennessee populations (**Fig. 3C**).

Among natural populations, we detected significant isolation by geographic distance (IBD) (Mantel's $r = 0.505$, $p = 0.001$). AMOVA further showed that genetic variation was partitioned across multiple hierarchical levels, with the partitioning of genetic variation similar regardless of whether populations were grouped into two, three, or four groups (**Table 2**). When populations were divided into two groups, 16% of the variation occurred between regions (IN+KY, TNE, and TNW), 23% among populations within regions, and 62% within populations. When populations were divided into three groups, 15% of the variation occurred between regions (IN+KY, TNE+TNW), 22% among populations within regions, and 63 % within populations.When populations were grouped into four groups, 15% of the variation occurred among regions (IN, KY, TNE, and TNW), 21% among populations within regions, and 64% within populations. The fact that two groups explained the greatest among-population variation is in agreement with the results of analyses of PCA and STRUCTURE.

Genetic diversity, inbreeding, and bottleneck

Populations of *P. globosa* varied in their genetic diversity (**Table 3**). The average number of alleles per locus (A) in *P. globosa* ranged from 2.278 – 6.5, the number of private alleles per population (A_P) ranged from 1-12, and the allelic richness (A_R) ranged from 2.14 – 4.52 (**Table 3**). *P. globosa* had slightly lower A_R than its rare sister species, *P. filiformis and P. ouachitensis*, and much lower A_R than the widespread species, *P. gracilis*. The *P. gracilis* population had an extraordinarily high number of private alleles (A_P) because we only sampled from one population. Within *P. globosa*, both private alleles per population (A_P) and the allele richness (A_R) were lowest in the population TNW_01 from western Tennessee (TNW), and greatest in

307 population KY_04 from Kentucky (KY) (**Fig. 4**). The translocated (TR) populations had
308 intermediate A_R but low A_P, which is consistent with genetic mixing of two or more populations.
309 In *Physaria globosa,* observed heterozygosity (H_O) ranged from 0.323 – 0.596, and
310 expected heterozygosity (H_E) ranged from 0.323 – 0.620, which is similar to *P. filiformis* and *P.*
311 *gracilis*, but greater than *P. ouachitensis.* Within *P. globosa*, most populations from Kentucky
312 (KY) exhibited greater than average H_O and H_E, and populations from both eastern Tennessee
313 (TNE) and western Tennessee (TNW) exhibited lower than average H_O and H_E (**Table 3**).
314 Generally, H_E was slightly greater than H_O in most populations, such that the average inbreeding
315 coefficient (F) was slightly positive (overall mean = 0.08, range -0.036 to 0.227) (**Table 3**).
316 However, these positive inbreeding coefficients are likely due in part to null alleles rather than
317 inbreeding, as the INEST analysis revealed that the percentage of null alleles (% null) ranged
318 from 0.0771 to 0.4694 (overall mean = 0.2648) (**Table 3**). After correcting for null alleles,
319 inbreeding coefficients (F_B) were generally very close to zero (range 0 – 0.1481; overall mean =
320 0.04; Table 3), indicating that most populations have not experienced inbreeding. However,
321 three populations (TNW_04, TNW_03, and TNW_01), all from the TNW region, exhibited
322 inbreeding coefficients greater than >0.10 even when corrected for null alleles, suggesting that
323 they may have experienced some inbreeding.
324 We used HP-rare to test for genetic bottlenecks and considered populations to show
325 significant evidence for a genetic bottleneck when three out of the four tests were statistically
326 significant. This revealed that the following six populations, representing three geographic
327 regions, had undergone recent bottlenecks: populations KY_02, KY_05, TR_02 from the KY
328 region; population TNE_01 from the TNE region; and populations TNW_01 and TNW_02 from

TNW region (**Fig. 4**). Generally, populations with the lowest A_R also exhibited significant evidence of either genetic bottlenecks (TNE_01, TNW_01, TNW_02), inbreeding (TNW_01 and TNW_03), or both (TNW_01).

Discussion

Our results indicate that conservation planning for *Physaria globosa* must balance two genetic realities: the species retains strong regional genetic structure that warrants conservation, but genetic vulnerability is unevenly distributed and may require population-specific interventions. *Physaria globosa* is neither genetically homogeneous across its range nor uniformly genetically depauperate across populations. Instead, the species exhibits strong regional differentiation and substantial variation in within-population diversity, with some populations displaying extremely low levels of genetic diversity and recent bottlenecks. The translocated populations exhibit clear genetic mixing, indicating that mixing material from different source populations can successfully produce viable offspring and increase genetic variation. However, the fitness consequences of genetic mixing remain unknown for this species and should be carefully evaluated. Taken together, these findings suggest that a multi-pronged conservation strategy involving protection of natural regional genetic structure, *ex situ* conservation of remaining genetic variation, and mixed-source translocations may help maintain genetic variation in *P. globosa*.

Genetic diversity and structure in *P. globosa*

Physaria globosa exhibits strong genetic structure across its disjunct geographic range, a pattern similar to that reported for the closely related species *Physaria filiformis* (C. E. Edwards et al. 2021). Neighbor-joining analysis, PCA, STRUCTURE, and pairwise genetic differentiation

consistently support a major split between the IN–KY and TNE–TNW groups, with additional substructure within each group. This pronounced spatial structure is likely reinforced by discrete habitats and limited dispersal. Seed movement in *P. globosa* appears to be largely gravity-mediated, and pollen dispersal may also be spatially restricted by localized pollinator activity involving bees, flies, and other invertebrates (Waters, Hurt, & Krosnick 2025). In addition, facultative self-compatibility in *P. globosa* may promote elevated inbreeding within populations (Powell 2023), further contributing to localized genetic structure. Significant isolation-by-distance and AMOVA results are consistent with this interpretation, indicating that genetic differentiation is structured both among regions and among populations within regions.

However, genetic structure was not fully explained by geographic distance alone. Two Tennessee populations (TNE_02 and TNW_03) were genetically similar despite occurring on opposite sides of the greater Nashville metropolitan area (**Fig. 2**). Given that these populations both occur along the Cumberland River, their similarity may reflect rare dispersal events via hydrological processes, flooding, animal vectors, human activity, or transportation corridors. For example, historical connectivity via the Tennessee Central Railway may have facilitated low levels of gene flow or movement of propagules (Hicks 2023). Thus, even though *P. globosa* populations are strongly genetically structured overall, occasional dispersal or historical connectivity may contribute to genetic similarity among geographically distant populations.

Genetic diversity in *P. globosa* was unevenly distributed across its range. Allelic richness (A_R) was generally lower in *P. globosa* than in its congeners, but diversity also varied substantially among populations. Generally, diversity was greater in the northern part of the species' range in IN and KY, whereas genetic diversity was lower in the TN populations. For

example, KY_04 exhibited the greatest allelic richness and number of private alleles, whereas TNW_01 had the lowest. Bottleneck and inbreeding analyses support this pattern, indicating that genetic risk is localized in the southern portion of the species' range rather than pervasive across the species. Spatial variation in genetic diversity may, in part, reflect differences in disturbance and management histories. In the southern portion of the range, construction of locks and dams along the Cumberland River (Olson 2021), historical railroad development (Hicks 2023), roadside herbicide use, recreation, and development (Long et al. 2021) may have reduced population sizes and altered connectivity in TN. Invasive *Lonicera maackii* may further contribute to local genetic erosion by reducing light availability and decreasing native plant growth (McNeish & McEwan 2016), a threat particularly relevant for a species potentially dependent on high-light conditions maintained by periodic disturbance (Krosnick et al. 2022; Long et al. 2021). Conversely, KY_04, which has been regularly managed with *L. maackii* removal since 2018, exhibited among the highest genetic diversity of all sampled populations. Although our data cannot directly test whether these management practices promoted high diversity, this pattern is consistent with the hypothesis that active habitat management may promote recruitment and help maintain genetic variation, possibly through contributions from a persistent soil seed bank.

Genetic variation in mixed-source translocations

Genetic mixing of multiple source populations can be used in *ex situ* and translocated populations to reduce inbreeding and increase gene flow and genetic diversity, but such intervention should be guided by regional genetic structure, source-population history, local adaptation, and post-translocation monitoring rather than treated as universally beneficial (Van

Rossum et al. 2025; Van Rossum & Le Pajolec 2021). In *Physaria globosa*, this concern is particularly relevant because genetic mixing and translocation efforts have been conducted for decades without prior genetic assessment or long-term genetic monitoring. This study documented the genetic composition of two genetically mixed populations whose genetic sources originated across different geographic scales. One genetically mixed population (TR_01) was sourced from two populations within the same geographic region, where gene flow between the source populations could occur naturally. In contrast, the other genetically mixed population (TR_02) was sourced from two geographically distinct regions, between which natural gene flow rarely occurs.

Translocations that promote genetic mixing and increase genetic diversity may be less controversial when they restore or augment gene flow among populations that are naturally or historically connected, because such interventions are less likely to disrupt historical population structure or local adaptation, if present. The TR_01 population was initially established using seeds from the KY_02 population and was subsequently augmented with seeds from the KY_05 population. Based on our population structure analysis, natural gene flow has occurred between these two source populations. There is no sign of population bottleneck in the TR_01 population, suggesting that the founder population of 100-200 seeds from 30-40 maternal plants was sufficient to maintain a level of genetic diversity comparable to that observed in natural populations. When we sampled the TR_01 population in 2021, it had been 7 years since the introduction of the KY_02 population and 6 years since the introduction of the KY_05 population. However, we only detected that 16.7% of the population was admixed. The low proportion of admixture could be attributable in part to the facultative self-compatibility

417 exhibited by the species. However, it is likely mostly attributable to the mostly biennial life
418 history of the species. Given that more than 95% of *P. globosa* individuals exhibited a biennial
419 life history, flowering in their 2nd year in a greenhouse experiment (unpublished data),
420 individuals sourced from KY_02 and KY_05 and introduced in different years may have had
421 limited overlap in flowering phenology, thereby reducing opportunities for genetic mixing.
422 These findings suggest that flowering phenology may need to be taken into consideration for
423 managing reintroductions of biennial species. Specifically, if mixing of different source
424 populations is desired, then they may need to be introduced in the same year, or new seed
425 could be introduced in the same year that existing plants are flowering.
426 In contrast, although translocations that mix naturally isolated populations can restore
427 genetic diversity and alleviate inbreeding depression, they may also alter natural population
428 structure, demographic history, expression of genetic load, or local adaptation, if present (Bell
429 et al. 2019; Frankham 2015; Pérez-Pereira, Caballero, & García-Dorado 2022; Sexton, Strauss, &
430 Rice 2011). Such interventions should therefore be considered only after careful evaluation and
431 when the target species faces a high risk of extinction without intervention. Our genetic
432 analysis indicated that the TR_02 population is a mixture of genotypes from the IN–KY and
433 TNE–TNW regions, which are geographically distinct and experience little natural gene flow.
434 Although this result demonstrates that individuals from these geographically distinct regions
435 can produce viable offspring, for the reasons discussed above, *we do not recommend mixing*
436 *source populations from different regions without first evaluating offspring performance and*
437 *genetic compatibility in a controlled setting*. Moreover, this translocated population also
438 showed a significant recent bottleneck with a deficit of rare alleles at segregation sites, which

implies this population was founded with few individuals instead of through well-planned genetic management. Importantly, although some *P. globosa* populations have experienced demographic bottlenecks and reduced genetic diversity, substantial genetic diversity is still maintained within geographic regions, and there are no clear signs that the species faces imminent extinction.

Conservation recommendations

Our genetic findings support a risk-benefit framework for conservation management for *P. globosa*. Strong regional structure (IN-KY vs. TNE-TNW) argues against treating all populations as interchangeable seed sources, as such practices could erode historic genetic differentiation. Conversely, reduced diversity, bottleneck signals, localized inbreeding, and ongoing habitat threats suggest that current management practices alone may be insufficient in some cases. The most defensible interpretation is therefore not that management must choose between conserving population structure or restoring gene flow, but that both goals must be balanced, such that a combination of in situ and ex situ conservation may be optimal.

In situ conservation should remain the primary focus because it maintains the geographic distribution of genetic diversity and supports representation and redundancy across the range. Efforts should prioritize the protection and management of multiple populations within each of the major geographic regions (IN, KY, TNE, TNW). At present, Indiana (IN) may be the only region where this goal is largely met, because its single population is both protected and actively managed. In eastern Tennessee (TNE), several populations along the Cumberland River are protected, and populations associated with cliff-faces and ledges may be less vulnerable to invasion by *L. maackii*; however, the genetic status of many of the sites in this

region remains uncertain due to limited access during this study. Many populations in Kentucky (KY) and western Tennessee (TNW) occur on private land and are not currently being actively managed, making them important targets for future conservation. Populations showing low allelic richness, bottleneck signatures, elevated inbreeding, or severe habitat degradation should therefore be prioritized for closer demographic and genetic monitoring, habitat management, and, where appropriate, demographic or genetic augmentation. However, these efforts may be logistically challenging because *P. globosa* often occurs on cliff tops, ledges, and steep slopes that are difficult and potentially dangerous to access, making them important targets for *ex situ* conservation.

Where in situ conservation is difficult, e*x situ* conservation and translocations may be complementary approaches to conserve *P. globosa*. Whenever possible, we recommend that these efforts be guided by genetic data to ensure that only populations that have experienced recent gene flow are mixed and that these efforts are coupled with continued monitoring and evaluation to assess their success. One example of where this could be useful is in the population TNW_01, which showed the lowest allelic richness and heterozygosity among all studied populations. If future studies show that TNW_01 has experienced rapid population decline and strong inbreeding depression, we would suggest conducting genetic rescue by translocating individuals from TNW_04 or TNW_05, which are closely related but have higher genetic diversity, into TNW_01.

Finally, this study highlights the limitation of inference based solely on neutral microsatellite markers. Although these markers effectively characterize genetic structure, relative population genetic diversity, and admixture, they do not provide insight into adaptive

variation among regional groups or the fitness consequences of genetic mixing for translocation. This uncertainty is particularly relevant for a species in which conservation decisions may involve translocating genotypes among historically differentiated populations. Future work integrating demographic data, common-garden or fitness experiments, and genome-scale markers will be necessary to determine whether management interventions and multi-source conservation translocations enhance long-term persistence and evolutionary potential in *P. globosa*.

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Reference list

Aguilar-Gómez, D., Yuan, L., Zhang, Y., Ochoa, A., Culver, M., Fitak, R. R., . . . Nielsen, R. (2025). Genetic rescue of Florida panthers reduced homozygosity but did not swamp ancestral genotypes. *Proceedings of the National Academy of Sciences, 122*(31), e2410945122. doi:doi:10.1073/pnas.2410945122

Allen, G. C., Flores-Vergara, M. A., Krasynanski, S., Kumar, S., & Thompson, W. F. (2006). A modified protocol for rapid DNA isolation from plant tissues using cetyltrimethylammonium bromide. *Nature Protocols, 1*(5), 2320–2325. doi:10.1038/nprot.2006.384

Bell, D. A., Robinson, Z. L., Funk, W. C., Fitzpatrick, S. W., Allendorf, F. W., Tallmon, D. A., & Whiteley, A. R. (2019). The exciting potential and remaining uncertainties of genetic rescue. *Trends in Ecology & Evolution, 34*(12), 1070–1079. doi:10.1016/j.tree.2019.06.006

Boutin-Ganache, I., Raposo, M., Raymond, M., & Deschepper, C. F. (2001). M13-tailed primers improve the readability and usability of microsatellite analyses performed with two different allele-sizing methods. *Biotechniques, 31*, 24–25.

509 Bucharova, A. (2017). Assisted migration within species range ignores biotic interactions and
510 lacks evidence. *Restoration Ecology, 25*(1), 14–18.
511 doi:https://doi.org/10.1111/rec.12457

512 Chybicki, I. J., & Burczyk, J. (2009). Simultaneous estimation of null alleles and inbreeding
513 coefficients. *Journal of Heredity, 100*, 106–113.

514 Donat, M. G., Lowry, A. L., Alexander, L. V., O'Gorman, P. A., & Maher, N. (2016). More extreme
515 precipitation in the world's dry and wet regions. *Nature Climate Change, 6*(5), 508–513.
516 doi:10.1038/nclimate2941

517 Dussex, N., Morales, H. E., Grossen, C., Dalen, L., & van Oosterhout, C. (2023). Purging and
518 accumulation of genetic load in conservation. *Trends in Ecology & Evolution, 38*(10),
519 961–969. doi:10.1016/j.tree.2023.05.008

520 Earl, D. A., & vonHoldt, B. M. (2012). STRUCTURE HARVESTER: a website and program for
521 visualizing STRUCTURE output and implementing the Evanno method. *Conservation*
522 *Genetics Resources, 4*(2), 359–361. doi:10.1007/s12686-011-9548-7

523 Edwards, C. E., Bassüner, B., David, A. S., Dell, N., Foye, C. J., Herron, S., . . . Menges, E. S.
524 (2025). Parentage analysis of conservation translocations in the endangered, self-
525 incompatible shrub *Pseudoziziphus celata* reveals reproductive constraints, variable
526 survival and a case for genetics-driven conservation. *Annals of Botany*.
527 doi:10.1093/aob/mcaf259

528 Edwards, C. E., Tessier, B. C., Swift, J. F., Bassüner, B., Linan, A. G., Albrecht, M. A., &
529 Yatskievych, G. A. (2021). Conservation genetics of the threatened plant species
530 *Physaria filiformis* (Missouri bladderpod) reveals strong genetic structure and a possible
531 cryptic species. *PLoS ONE, 16*, e0247586.

532 Elberling, H. (2000). Spatial pattern of *Lesquerella arctica*: effects of seed bank and desiccation
533 cracks. *Écoscience, 7*, 86–91.

534 Faircloth, B. C. (2008). MSATCOMMANDER: detection of microsatellite repeat arrays and
535 automated, locus-specific primer design. *Molecular Ecology Resources, 8*, 92–94.

536 *Flora of North America*. (2010). (Vol. 7). New York and Oxford: Oxford University Press.

537 Frankham, R. (2015). Genetic rescue of small inbred populations: meta-analysis reveals large
538 and consistent benefits of gene flow. *Molecular Ecology, 24*(11), 2610–2618.
539 doi:https://doi.org/10.1111/mec.13139

Fraser, D. J., & Bernatchez, L. (2001). Adaptive evolutionary conservation: towards a unified concept for defining conservation units. *Molecular Ecology, 10*(12), 2741–2752. doi:https://doi.org/10.1046/j.0962-1083.2001.01411.x

Fuentes-Soriano, S., & Al-Shehbaz, I. (2013). Phylogenetic relationships of mustards with multiaperturate pollen (Physarieae, Brassicaceae) based on the plastid *ndhF* gene: Implications for morphological diversification. *Systematic Botany, 38*(1), 178–191. doi:10.1600/036364413x661854

Fuentes-Soriano, S., & Kellogg, E. A. (2021). Molecular systematics of tribe Physarieae (Brassicaceae) based on nuclear *ITS, LUMINIDEPENDENS*, and chloroplast *ndhF*. *Systematic Botany, 46*(3), 611–627. doi:10.1600/036364421x16312067913318

Gaitán-Espitia, J. D., & Hobday, A. J. (2021). Evolutionary principles and genetic considerations for guiding conservation interventions under climate change. *Global Change Biology, 27*(3), 475–488. doi:https://doi.org/10.1111/gcb.15359

Gassert, F., Venter, O., Watson, J. E. M., Brumby, S. P., Mazzariello, J. C., Atkinson, S. C., & Hyde, S. (2023). An operational approach to near real-time global high-resolution mapping of the terrestrial Human Footprint. *Frontiers in Remote Sensing, Volume 4 - 2023*. doi:10.3389/frsen.2023.1130896

Hicks, J. A. (2023). A brief history of the Tennessee Central Railway Company. Retrieved from https://www.tcry.org/tcry-history

Hoelzel, A. R., Bruford, M. W., & Fleischer, R. C. (2019). Conservation of adaptive potential and functional diversity. *Conservation Genetics, 20*(1), 1–5. doi:10.1007/s10592-019-01151-x

Hufford, K. M., & Mazer, S. J. (2003). Plant ecotypes: genetic differentiation in the age of ecological restoration. *Trends in Ecology & Evolution, 18*(3), 147–155. doi:10.1016/S0169-5347(03)00002-8

Jombart, T. (2008). adegenet: a R package for the multivariate analysis of genetic markers. *Bioinformatics, 24*, 1403–1405.

Kalinowski, S. T. (2005). hp-rare 1.0: a computer program for performing rarefaction on measures of allelic richness. *Molecular Ecology Notes, 5*, 187–189.

Kallison, E. R., Hettihewa, B., Mashburn, B., O'Kane, S. L., Jr., Albrecht, M. A., & Edwards, C. E. (2026). Morphological and phylogenomic analyses support the recognition of a cryptic

new species of *Physaria* (Brassicaceae) endemic to the Ouachita Mountains of Arkansas, USA. *Systematic Botany*.

Krosnick, S. E., Thacker, J. H., Mattingly, H. T., Call, G. P., Maynord, S. C., Adams, D. S., & Wheeler, K. (2022). Ecological correlates of reproductive output in a Tennessee population of short's bladderpod, *Physaria globosa* (Brassicaceae). *Castanea, 87*, 20–38.

Linan, A. G., Gereau, R. E., Sucher, R., Mashimba, F. H., Bassuner, B., Wyatt, A., & Edwards, C. E. (2024). Capturing and managing genetic diversity in ex situ collections of threatened tropical trees: A case study in *Karomia gigas*. *Applications in Plant Sciences, 12*(3), e11589. doi:10.1002/aps3.11589

Long, Q., Albrecht, M., Call, G., & Region, S. (2021). *Recovery implementation strategy for short's bladderpod (Physaria globosa)*. Retrieved from https://ecos.fws.gov/docs/recovery_plan/20210805_P.globosa_RIS.pdf

MacDonald, Z. G., Dupuis, J. R., Glasier, J. R. N., Sissons, R., Moehrenschlager, A., Shaffer, H. B., & Sperling, F. A. H. (2025). Whole-genome evaluation of genetic rescue: The case of a curiously isolated and endangered butterfly. *Molecular Ecology, 34*(4), e17657. doi:https://doi.org/10.1111/mec.17657

Mashburn, B., Jhangeer-Khan, R., Bégué, A., Tatayah, V., Olsen, K. M., & Edwards, C. E. (2023). Genetic assessment improves conservation efforts for the critically endangered oceanic island endemic *Hibiscus liliiflorus*. *Journal of Heredity, 114*(3), 259–270. doi:10.1093/jhered/esad021

McNeish, R. E., & McEwan, R. W. (2016). A review on the invasion ecology of Amur honeysuckle (*Lonicera maackii*, Caprifoliaceae) a case study of ecological impacts at multiple scales. *The Journal of the Torrey Botanical Society, 143*, 367–385.

Meirmans, P. G., & Van Tienderen, P. H. (2004). GENOTYPE and GENODIVE: two programs for the analysis of genetic diversity of asexual organisms. *Molecular Ecology Notes, 4*, 792–794.

Olson, K. (2021). Cumberland River resource stewardship and protection: Managing the Cumberland River and the land between the lakes landscapes. *Journal of Water Resource and Protection, 13*, 92–111.

Paradis, E., Claude, J., & Strimmer, K. (2004). APE: Analyses of Phylogenetics and Evolution in R language. *Bioinformatics, 20*(2), 289–290. doi:10.1093/bioinformatics/btg412

Peakall, R., & Smouse, P. E. (2012). GenAlEx 6.5: genetic analysis in Excel. Population genetic software for teaching and research--an update. *Bioinformatics, 28*(19), 2537–2539. doi:10.1093/bioinformatics/bts460

Pérez-Pereira, N., Caballero, A., & García-Dorado, A. (2022). Reviewing the consequences of genetic purging on the success of rescue programs. *Conservation Genetics, 23*(1), 1–17. doi:10.1007/s10592-021-01405-7

Powell, E. D. (2023). *Ecology and reproductive dynamics of Physaria globosa (Brassicaceae).* (M.S.), Tennessee Technological University, United States -- Tennessee. ProQuest Dissertations & Theses Global database. (30485652)

Pritchard, J. K., Stephens, M., & Donnelly, P. (2000). Inference of population structure using multilocus genotype data. *Genetics, 155*(2), 945–959. doi:10.1093/genetics/155.2.945

Rozen, S., & Skaletsky, H. (1999). Primer3 on the WWW for general users and for biologist programmers. *Methods Molecular Biology, 132*, 365–386.

Sexton, J. P., Strauss, S. Y., & Rice, K. J. (2011). Gene flow increases fitness at the warm edge of a species' range. *Proceedings of the National Academy of Sciences, 108*(28), 11704–11709. doi:10.1073/pnas.1100404108

Shaw, R. E., Farquharson, K. A., Bruford, M. W., Coates, D. J., Elliott, C. P., Mergeay, J., . . . Grueber, C. E. (2025). Global meta-analysis shows action is needed to halt genetic diversity loss. *Nature, 638*(8051), 704–710. doi:10.1038/s41586-024-08458-x

Swift, J. F., Smith, S. A., Menges, E. S., Bassuner, B., & Edwards, C. E. (2016). Analysis of mating system and genetic structure in the endangered, amphicarpic plant, Lewton's polygala (*Polygala lewtonii*). *Conservation Genetics*, 1–16.

Van Rossum, F., Buisson, D., Le Péchon, T., Daco, L., Hermant, S., Dahlem, R., . . . Bardin, P. (2025). Taxonomic and genetic assessment of disjunct populations of the threatened plant *Arnica montana* (Asteraceae) from central France. *Perspectives in Plant Ecology, Evolution and Systematics, 67*, 125873. doi:https://doi.org/10.1016/j.ppees.2025.125873

Van Rossum, F., & Le Pajolec, S. (2021). Mixing gene pools to prevent inbreeding issues in translocated populations of clonal species. *Molecular Ecology, 30*(12), 2756–2771. doi:https://doi.org/10.1111/mec.15930

631 Waters, C., Hurt, C., & Krosnick, S. (2025). Looking to the past to inform the future: What eDNA
632 from herbarium specimens can tell us about plant–animal interactions. *Applications in*
633 *Plant Sciences, 13*(2), e11633. doi:https://doi.org/10.1002/aps3.11633

634 Xu, C., McDowell, N. G., Fisher, R. A., Wei, L., Sevanto, S., Christoffersen, B. O., . . . Middleton, R.
635 S. (2019). Increasing impacts of extreme droughts on vegetation productivity under
636 climate change. *Nature Climate Change, 9*(12), 948–953. doi:10.1038/s41558-019-0630-
637 6

638 Yu, G. (2025). scatterpie: Scatter Pie Plot. Retrieved from https://github.com/yulab-
639 smu/scatterpie

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Statements & Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

W.-H. K. analyzed the data, visualized the results, and wrote the manuscript. C. L. analyzed the data and wrote the manuscript. B. B. conducted molecular experiments. A. G. L. conducted fieldwork, collected samples, and revised the manuscript. T. R. L. conducted fieldwork, collected samples, and wrote the manuscript. M. A. A. conceived the project, conducted fieldwork, collected samples, and revised the manuscript. C. E. E. conceived of the project, obtained funding, q ap0-1q2conducted fieldwork, collected samples, analyzed the data, and wrote the manuscript. All authors read and approved the final manuscript.

Data availability

The datasets generated and analyzed during the current study are available in the Physaria_SSR repository, https://github.com/kuowenhsi/Physaria_SSR.

Tables

Table 1. Information about the populations and samples of *Physaria globosa* included in the study, including voucher number, collection locality, ecoregion, GPS coordinates, and number of samples collected per site.

Population	Collector¶	Species	State, County, Locality (Element Occurrence number)	Ecoregion	Latitude	Longitude	Number of samples
IN_01	Edwards and Linan 346 (MO)	*P. globosa*	IN, Posey, Big Cypress Slough	Shawnee Hills	37.82	-88.00	24
KY_01	Edwards, Linan and Littlefield 349 (MO)	*P. globosa*	KY, Franklin. Near intersection of Camp Pleasant Rd. and Gregory Woods Rd. (EO-23a)	Bluegrass	38.31	-84.80	24
KY_02	Edwards, Linan and Littlefield 347 (MO)	*P. globosa*	KY, Franklin, Cove Spring. Area north of parking lot on 127, just north of its intersection with 421 (EO18)	Bluegrass	38.22	-84.85	11
KY_03	Edwards, Linan and Littlefield 350 (MO)	*P. globosa*	KY, Franklin. Bald Knob (EO-07)	Bluegrass	38.20	-84.88	23
KY_04	Edwards, Linan and Littlefield 351 (MO)	*P. globosa*	KY, Franklin. Ninevah Rd. (EO-04)	Bluegrass	38.11	-84.87	24
KY_05	Edwards, Linan and Littlefield 352 (MO)	*P. globosa*	KY, Woodford. Off Delaney Ferry Rd. (Murphy property, EO-28)	Bluegrass/ partially cultivated	37.97	-84.69	24
KY_06	Edwards, Linan and Littlefield 354 (MO)	*P. globosa*	KY, Madison. Valley view on bluff over Tate Creek. In thick honeysuckle (EO10a)	Bluegrass	37.84	-84.42	11
TR_01	Edwards, Linan and Littlefield 348 (MO)	*P. globosa*	KY, Franklin. Cove Spring (EO30). Source: KY_02 and KY_05	Translocation	38.22	-84.84	24
TR_02	Edwards, Linan and Littlefield 353 (MO)	*P. globosa*	KY, Jessamine. Jessamine Creek Gorge Preserve, at the entrance to the parking area	Translocation	37.83	-84.64	24
TNE_01	Edwards, Albrecht, and Elam 357 (MO)	*P. globosa*	TN, Trousdale. East of Hartsville Boat Ramp (EO03)	Nashville Basin	36.38	-86.17	24
TNE_02	Edwards, Albrecht, and Elam 356 (MO)	*P. globosa*	TN, Smith. North of Cordell Hull Dam, bluffs over Cumberland River (EO24)	Nashville Basin	36.31	-85.94	24
TNW_01	Edwards, Linan and Dell 362 (MO)	*P. globosa*	TN, Montgomery. King and Queen's Bluff (EO 12)	Highland Rim	36.50	-87.32	14
TNW_02	Edwards, Linan and Dell 368 (MO)	*P. globosa*	TN, Montgomery. Jarrell Ridge Road (EO28)	Highland Rim	36.39	-87.26	9

TNW_03	Edwards, Linan and Dell 367 (MO)	*P. globosa*	TN, Cheatham. Lock B Rd., near a recycle center/train tracks (EO01)	Highland Rim	36.24	-87.04	24
TNW_04	Edwards, Linan and Dell 365 (MO)	*P. globosa*	TN, Davidson. McMinn's bluff (EO04)	Highland Rim	36.21	-86.98	24
TNW_05	Edwards, Linan and Dell 366 (MO)	*P. globosa*	TN, Davidson/Cheatham. McMinn's bluff (EO10)	Highland Rim	36.21	-86.96	24
AR_01*	Yatskievych 15-003 (MO)	*P. ouachitensis*	AR, Garland. Cedar Fourche Recreation Area	Ouachita mountains	34.67	-93.28	5
AR_02*	Yatskievych 15-009 (MO)	*P. ouachitensis*	AR, Garland. North side of Lake Ouachita	Ouachita mountains	34.66	-93.27	9
AR_03*	Yatskievych 15-012 (MO)	*P. ouachitensis*	AR, Hos Spring. Ross Foundation property	Ouachita mountains	34.35	-93.07	4
MO_06*	Rimer s.n. (MO)	*P. filiformis*	MO, Greene. Seepside Glade	Ozark Plateau	37.30	-93.41	4
MO_10*	Edwards 281	*P. filiformis*	MO, Greene. Wilsons Creek, Wire Road	Ozark Plateau	37.09	-93.42	6
MO_11*	Edwards 282	*P. filiformis*	MO, Greene. Wilsons Creek, Bloody Hill	Ozark Plateau	37.10	-93.41	7
TX_01*	Yatskievych 16-110 (TEX)	*P. gracilis*	TX, Williamson. Near San Gabriel River	Coastal Plains	30.66	-97.46	11

* Samples collected as part of a previous study on *Physaria filiformis* (Edwards et al. 2021)
¶ Links to Tropicos system if available

Table 2. AMOVA summary table with the Indiana (IN), Kentucky (KY), Tennessee east (TNE), and Tennessee west (TNW) populations divided into 1) two groups (IN+KY, TNE+TNW), 2) three groups (IN+KY, TNE, TNW), or 3) the four ecoregions they occupy: Indiana (IN), Kentucky (KY), Tennessee east (TNE), and Tennessee west (TNW),

Source of variation	df	SS	MS	Est. Var. (%)	F-statistic
2 regions (IN+KY, TNE+TNW)					
Among Regions	1	412.943	412.943	1.205 (16%)	F_{RT}=0.156
Among Pops	12	873.469	72.789	1.743 (23%)	F_{SR}=0.267
Within Pops	540	2584.181	4.786	4.786 (62%)	F_{ST}=0.381
Total	553	3870.592		7.733 (100%)	100%
3 regions: (IN+KY, TNE, TNW)					
Among Regions	2	532.768	266.384	1.104 (15%)	F_{RT}=0.146
Among Pops	11	753.643	68.513	1.658(22%)	F_{SR}=0.257
Within Pops	540	2584.181	4.786	4.786 (63%)	F_{ST}=0.366
Total	553	3870.592		7.548 (100%	
4 regions (IN, KY, TNE, TNW)					
Among Regions	3	640.363	213.454	1.084 (15%)	F_{RT}=0.145
Among pops within regions	10	646.048	64.605	1.591 (21%)	F_{SR}=0.245
Within populations	540	2584.181	4.786	0.825 (64%)	F_{ST}=0.358
Total	553	3870.592		7.459 (100%)	

Table 3. Genetic diversity metrics for *Physaria globosa* and related species.

Population	Region	n	A	A_R	A_P	Ho	He	F_{IS}	% null	F^B
P. globosa										
TR_01	KY	24	4.72	3.73	1	0.48	0.53	0.08	0.46	0.06
TR_02	KY	24	4.39	3.54	1	0.55	0.62	0.09	0.18	0.07
IN_01	IN	24	3.28	2.82	2	0.4	0.45	0.08	0.16	0
KY_01	KY	22	4.11	3.43	7	0.55	0.56	0	0.22	0
KY_02	KY	10	4	3.74	1	0.5	0.51	0.05	0.18	0
KY_03	KY	20	5.33	4.1	4	0.56	0.6	0.07	0.3	0
KY_04	KY	24	6.5	4.52	12	0.6	0.62	0.03	0.26	0
KY_05	KY	24	3.94	3.32	1	0.51	0.53	0.03	0.18	0.05
KY_06	KY	11	4.06	3.7	4	0.53	0.55	0.06	0.28	0
TNE_01	TNE	23	3.22	2.64	10	0.38	0.41	0.11	0.15	0
TNE_02	TNE	24	4.17	3.23	9	0.41	0.43	0.02	0.33	0
TNW_01	TNW	14	2.28	2.14	1	0.32	0.32	0.03	0.08	0.14
TNW_02	TNW	9	2.78	2.69	1	0.48	0.46	-0.04	0.17	0
TNW_03	TNW	24	3.28	2.69	3	0.37	0.44	0.23	0.47	0.11
TNW_04	TNW	24	4.72	3.62	6	0.4	0.5	0.16	0.33	0.15
TNW_05	TNW	24	5.44	3.87	6	0.48	0.55	0.11	0.36	0.07
Average across *P. globosa*			4.2	3.38	4.24	0.47	0.51	0.08	0.26	0.04
P. ouachitensis										
AR_01	AR	5	1.79	7.14	3	0.17	0.27	0.37	0.45	0
AR_02	AR	9	2.4	5.52	2	0.16	0.3	0.46	0.53	0.38
AR_03	AR	4	1.85	6.37	2	0.15	0.4	0.61	0.57	0
Average across *P. ouachitensis*			2.01	6.34	2.33	0.16	0.32	0.48	0.52	0.13
P. filiformis										
MO_06	MO	4	2.53	5.95	4	0.45	0.49	0.08	0.24	0
MO_10	MO	6	3.18	4.38	8	0.42	0.61	0.32	0.54	0
MO_11	MO	7	2.81	4.25	5	0.35	0.48	0.29	0.42	0
Average across *P. filiformis*			2.84	4.86	5.67	0.4	0.53	0.23	0.4	0
P. gracilis										
TX_01	TX	11	5	8.42	42	0.534	0.711	0.249	0.77	0

Column definitions: n=number of individuals; A=average alleles per locus; A_R=allelic richness; A_P=private alleles per population; H_O=observed heterozygosity; H_E=expected heterozygosity; F=fixation index; % null=percentage of null alleles; F_B=fixation index adjusted for null alleles.

Figures

A



B

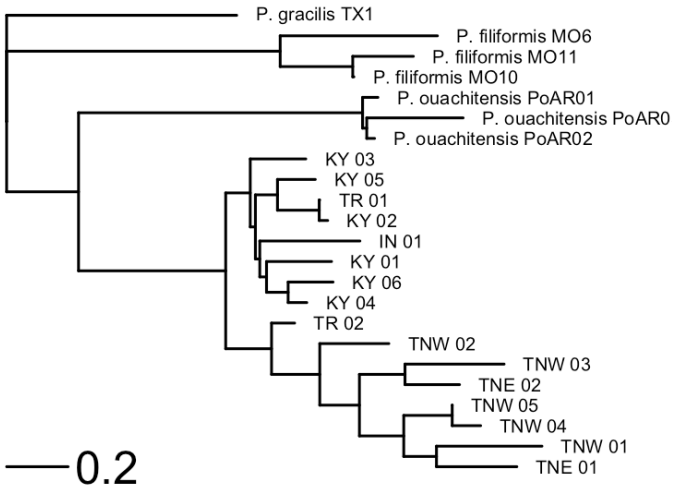


Figure 1. Photos of *Physaria globosa* and genetic distance tree.
(**A**) Native habitat. (**B**) Neighbor-Joining tree based on Nei's genetic distances. The tree was rooted at *P. gracilis*.

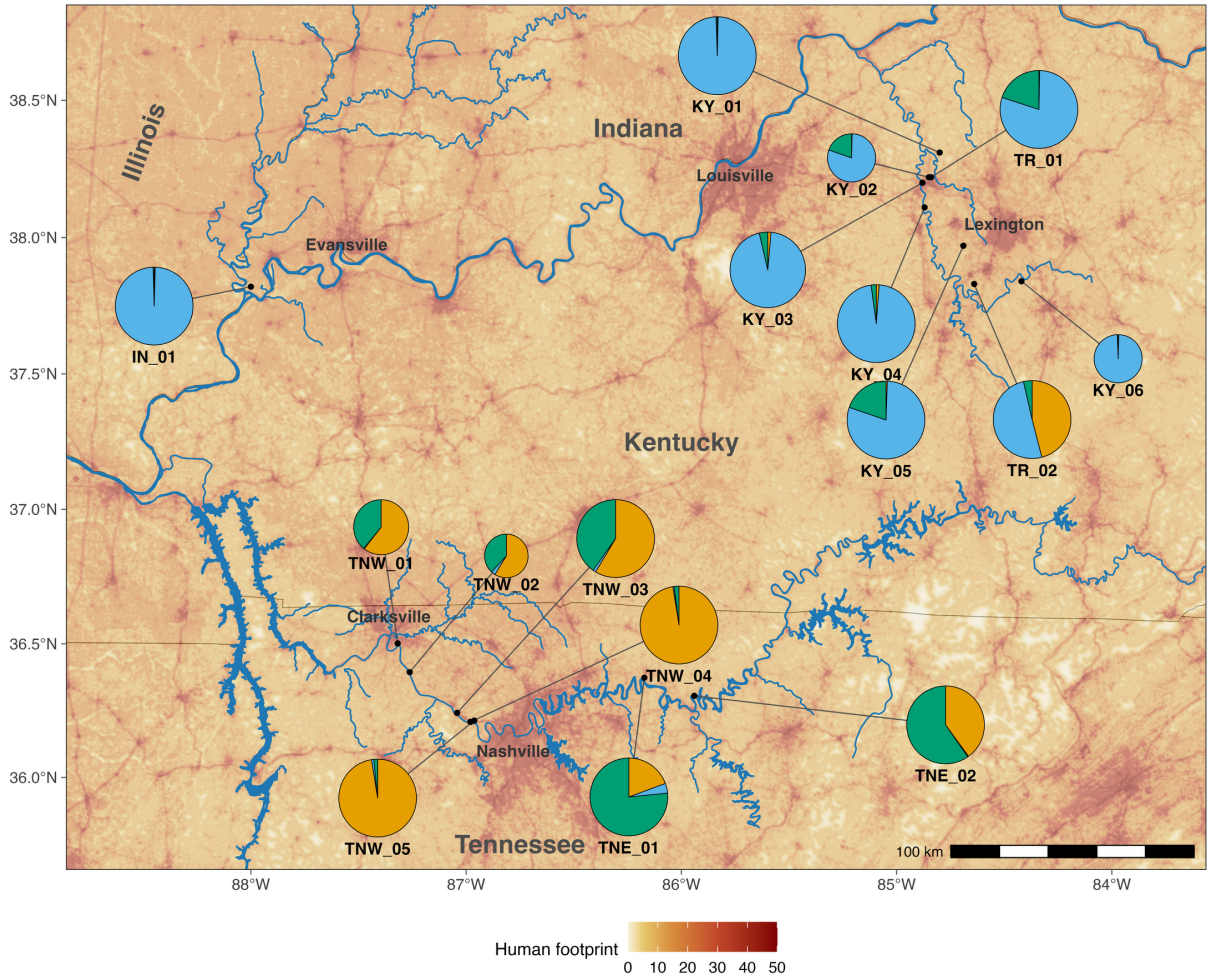


Figure 2. Map of *P. globosa* collection localities, with population structure K = 3 shown in pie charts.

The size of the pie chart reflects the sampling size per population. See **Table 1** for detailed information.

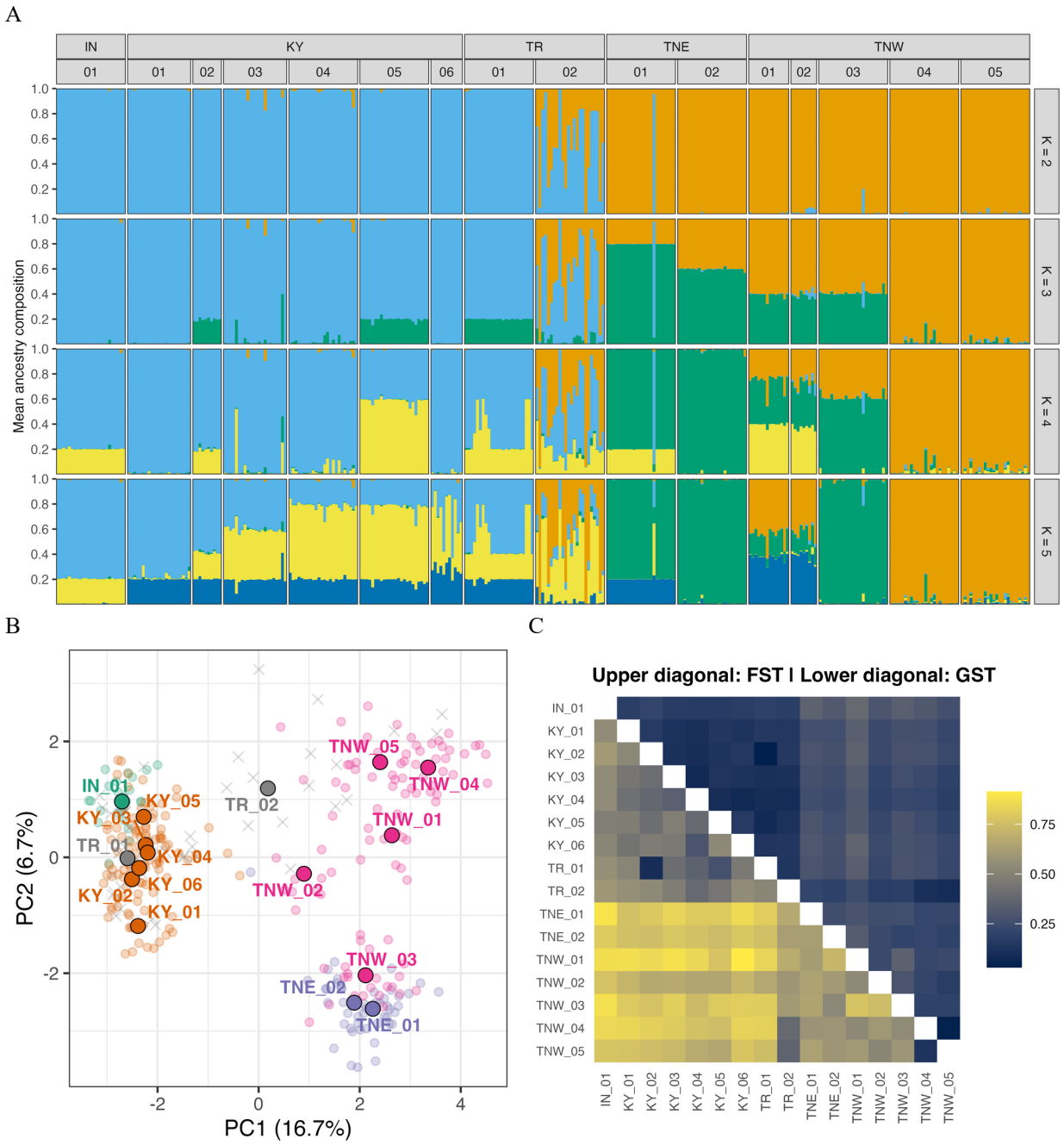


Figure 3. PCA of all natural *P. globosa* populations, with location and species indicated. (A) STURCTURE analysis result (K = 2-5). (B) PCA plot. The big points with black stroke outline are population means. The small points without black stroke outline are individuals from natural populations. The crosses are individuals from *ex situ* populations. Points are colored by

regions and population codes are labeled. (C) heatmap of F_{ST} and G_{ST}. See **Table 1** for the population codes.

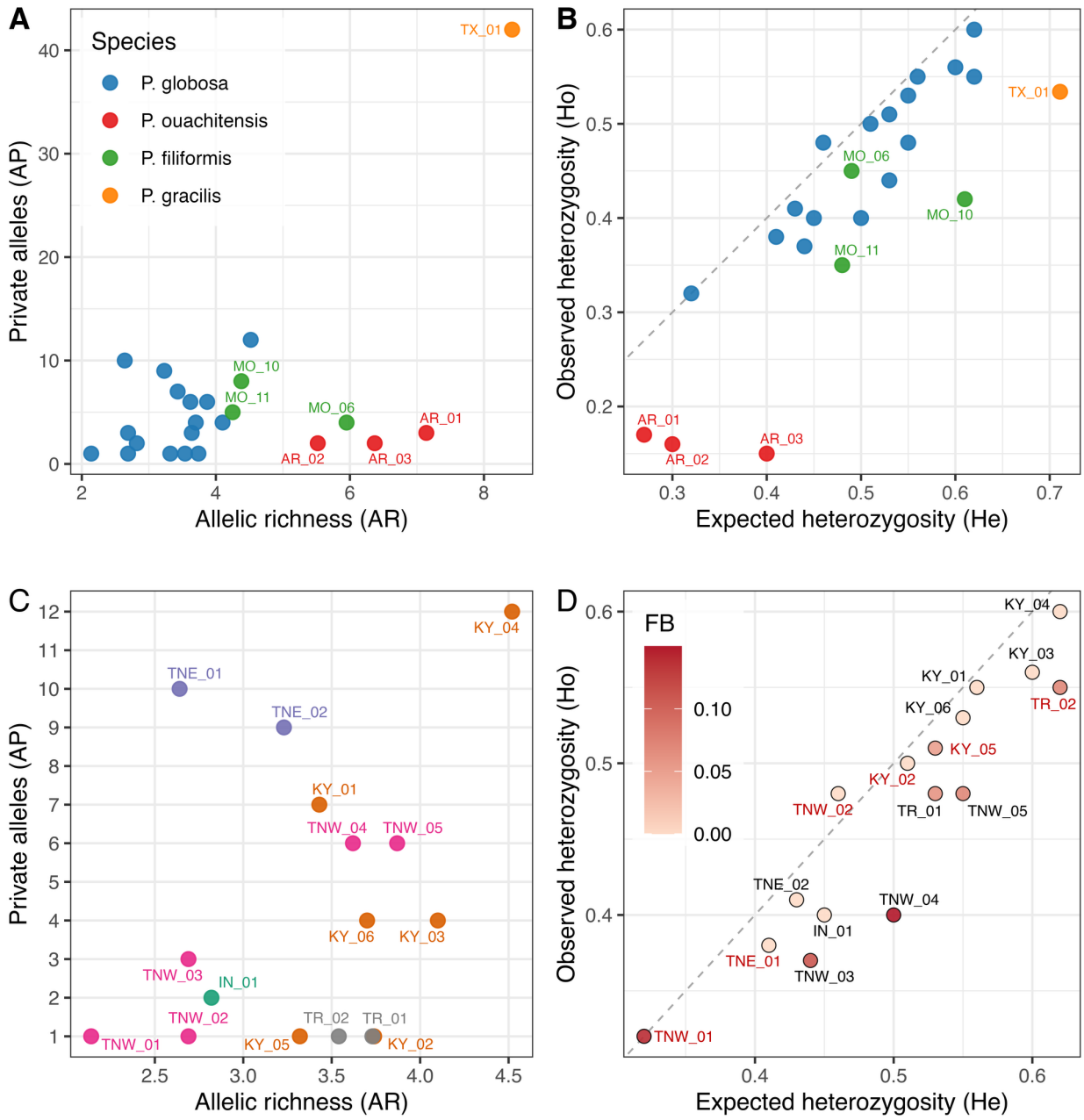


Figure 4. Genetic diversity analyses plots.

(**A**-**B**), *Physaria globosa* and outgroup species. (**C**-**D**), *Physaria globosa* with population labels. F_B., inbreeding coefficient after correction for null alleles. The six populations with excessive heterozygosity are labeled in red in (**D**). See **Table 1** for the population codes.